

HW 1: Confidence Intervals for Proportions

For the following table, fill in the missing information using the clues in the table.

Problem	Point Estimate	Sample Size	Confidence Level	Standard Error	Confidence Interval
1	$\hat{p} = 0.55$	$n = 150$	C-Level: 95% $z^* = 1.96$	$SE_{\hat{p}} = \underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$ $\underline{\hspace{2cm}}$
2	$\hat{p} = \underline{\hspace{2cm}}$	$n = 40$	C-Level: $\underline{\hspace{2cm}}$ $z^* = \underline{\hspace{2cm}}$	$SE_{\hat{p}} = \underline{\hspace{2cm}}$	$0.25 \pm 1.75(\underline{\hspace{2cm}})$ $\underline{\hspace{2cm}}$
3	$\hat{p} = \underline{\hspace{2cm}}$	$n = 80$	C-Level: $\underline{\hspace{2cm}}$ $z^* = 1.79$	$SE_{\hat{p}} = \underline{\hspace{2cm}}$	$\underline{\hspace{2cm}} \pm 1.79(\underline{\hspace{2cm}})$ $(0.42, 0.62)$
4	$\hat{p} = \underline{\hspace{2cm}}$	$n = \underline{\hspace{2cm}}$	C-Level: $\underline{\hspace{2cm}}$ $z^* = \underline{\hspace{2cm}}$	$\sqrt{\frac{0.70(1 - 0.70)}{120}}$ $SE_{\hat{p}} = 0.0418$	$\underline{\hspace{2cm}}$ $(0.6164, 0.7836)$

5) At the Renaissance Festival, there are many people who are dressed up in various characters. You are interested in the true proportion of people who dress-up for the festival. You randomly select 300 people who enter the festival on a randomly selected Saturday of the festival, and record whether they are dressed in a costume. You find that 125 of people have costumes on.

a) Explain the difference between p and $\hat{p}$ in the context of this problem?

b) Are the conditions needed for inference met? Explain.

c) You want to construct a 93% confidence interval to capture the true proportion of people who dress up in costume at the Renaissance Festival. Find the following: z^* , $SE_{\hat{p}}$, and $\hat{p}$

d) Find the 93% confidence interval and interpret the confidence interval in the context of the problem.